

DIFFERENTIATION_AUDIT.md — Core Concept Differentiation: Spec vs. Presentation

This document systematically audits the **18 signature concepts of ASSURE-TWIN** to determine whether the presentation clearly communicates them or leaves them unexploited.

The 18 Signature Concepts Audit Matrix

#	Core Concept	Implemented in Project?	Backend	Frontend	Demo Ready?	Mentioned in PPT?	Clarity in PPT
1	Virtual Downhole State	YES	analytics/state_adapter.py	WellStateView.ts	YES	PARTIAL	Poor
2	Dynamic Operating Envelope	YES	analytics/envelope.py	WellStateView.ts	YES	PARTIAL	Weak
3	Pumpability Window	YES	analytics/pumpability.py	WellStateView.ts	YES	NO	Unmentioned
4	Thermal Memory	YES	analytics/thermal_memory.py	WellStateView.ts	YES	NO	Unmentioned
5	Future Trajectory Forecast	YES	forecast/service.py	ForecastView.ts	YES	PARTIAL	Vague
6	Decision Rehearsal Sandbox	YES	forecast/rehearsal.py	ScenariosView.ts	YES	NO	Unmentioned
7	Candidate Robustness Scoring	YES	optimization/optimizer.py	OptimizationView.ts	YES	NO	Unmentioned

#	Core Concept	Implemented in Project?	Backend	Frontend	Demo Ready?	Mentioned in PPT?	Clarity in PPT
8	Economic & Tariff Sensitivity	YES	forecast/sensitivity.py	ScenariosView.ts	YES	NO	Unmentioned
9	Recommendation Contract	YES	api/v1/recommendations.py	RecommendationView.ts	YES	PARTIAL	Vague
10	The Abstain Protocol	YES	assurance/gatekeeper.py	AssuranceView.ts	YES	NO	Unmentioned
11	12-Point Assurance Gate	YES	assurance/gatekeeper.py	AssuranceView.ts	YES	PARTIAL	Weak
12	OOD Mahalanobis Detection	YES	ai/surrogate.py	N/A	YES	NO	Unmentioned
13	Physics vs. ML Agreement	YES	ai/model_agreement.py	AssuranceView.ts	YES	PARTIAL	Weak
14	Data Provenance Tracking	YES	models/entities.py	Workstation Badges	YES	NO	Unmentioned
15	Data Quality Screening	YES	calibration/sensor_screener.py	CalibrationView.ts	YES	PARTIAL	Generic
16	Outcome Reconciliation	YES	calibration/reconciler.py	CalibrationView.ts	YES	NO	Unmentioned

#	Core Concept	Implemented in Project?	Backend	Frontend	Demo Ready?	Mentioned in PPT?	Clarity in PPT
17	Interactive Lab Calibration	YES	<code>calibration/viscosity_fitter.py</code>	<code>CalibrationView.ts</code>	YES	PARTIAL	Generic
18	Immutable Audit Trail	YES	<code>models/entities.py</code>	<code>SettingsView.ts</code>	YES	PARTIAL	Brief

Summary Verdict on Differentiation

The current presentation **drastically sells the software short**:

- The project implements **18 sophisticated, production-grade differentiators**.
- The presentation **completely omits 7 of them** (including the two most powerful: **The Abstain Protocol** and **Decision Rehearsal Sandbox**).
- The presentation **weakly or generically explains 8 of them**.
- By replacing fictional claims (like downhole electric heaters) with these **authentic, fully implemented differentiators**, the presentation's technical credibility will surge.